

Holistic Data Preparer - Theory Concepts & Definitions

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Theory Concepts & Definitions

Project: Customer Credit Risk - Data Preprocessing & Feature Engineering **Duration:** 8 Days (1 hour/day)

Part A: Conceptual Foundation

Data Analysis is the process of inspecting, cleaning, transforming, and modeling data to discover useful information, draw conclusions, and support decision-making.

Planning a Data Science project typically follows these steps: understand the business problem → collect data → explore & clean → engineer features → model → communicate results → iterate.

Framing an ML problem means converting a business goal into a precise task: choosing the problem type (e.g. binary classification), defining inputs/outputs, and picking the right evaluation metric. Here the task is **binary classification** - predicting `default_flag` (0 = No Default, 1 = Default).

Tensor: a generalization of scalars (rank 0), vectors (rank 1), and matrices (rank 2) to n dimensions. A tabular dataset (rows x columns) is a rank-2 tensor.

Part B: Data Acquisition

- **CSV (Comma-Separated Values):** a plain-text tabular format, read with `pandas.read_csv()`.
 - **JSON (JavaScript Object Notation):** a nested key-value format, common for APIs and metadata, read with the `json` module or `pandas.read_json()`.
 - **SQL:** Structured Query Language used to query relational databases; here simulated with SQLite and read via `pandas.read_sql()`.
 - **API (Application Programming Interface):** a way for programs to request data from an external service, typically returning JSON.
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Part C: Data Understanding & Cleaning

- **`.info()`:** shows column dtypes, non-null counts, and memory usage.
- **`.describe()`:** shows summary statistics (mean, std, min, max, quartiles) for numeric columns.
- **Data Profiling:** an automated summary of a dataset's structure, quality, and distributions (missing %, cardinality, correlations).

Missing Data Handling Techniques

Technique	Definition
Simple Imputer (mean/median)	Replaces missing numeric values with the column's mean or median.
Simple Imputer (most frequent)	Replaces missing categorical values with the most common category.
Missing Indicator + Random Sample Imputation	Adds a binary flag for "was missing" and fills gaps by randomly sampling from observed values.
KNN Imputer	Fills missing values using the average of the k nearest neighboring rows (multivariate).
MICE (Multivariate Imputation by Chained Equations)	Iteratively models each column with missing values as a function of the other columns, cycling until convergence.
Complete Case Analysis	Drops all rows (or columns) containing any missing values.

Part D: Outlier Handling

Technique	Definition
Z-score Method	Flags a value as an outlier if $ z = (x - \text{mean}) / \text{std} > 3$ (commonly).
IQR Method	Flags values outside $[Q1 - 1.5 \times \text{IQR}, Q3 + 1.5 \times \text{IQR}]$, where $\text{IQR} = Q3 - Q1$.
Percentile Method	Flags values below the 1st percentile or above the 99th percentile.
Winsorization	Caps extreme values at a chosen percentile instead of removing them, preserving row count.

Part E: Feature Engineering

- **Mixed Variables:** columns containing both numeric and categorical (or multiple types within one field) that need type-aware handling.
- **Date/Time Feature Extraction:** deriving Year, Month, Day, Weekday, etc. from a raw date/timestamp column so models can use it.

Encoding Categorical Variables

Technique	Definition
Ordinal Encoding	Assigns integers to categories that have a natural order (e.g. Primary < Secondary < Graduate).

Label Encoding	Assigns a unique integer to each category, with no implied order - best for binary/target labels.
One-Hot Encoding	Creates a separate binary (0/1) column for each category - for nominal (unordered) variables.

Encoding Numerical Features

Technique	Definition
Binning	Groups continuous values into discrete intervals/buckets (e.g. income bands).
Binarization	Converts a numeric value into 0/1 based on a threshold.
Quantile Binning	Creates bins with (approximately) equal numbers of observations.
K-Means Binning	Uses K-Means clustering to find data-driven bin boundaries.

Part F: Feature Scaling

Technique	Definition
Standardization (Z-score scaling)	Transforms data to mean 0, standard deviation 1: $z = (x - \text{mean}) / \text{std.}$
Normalization	Scales each row (sample) to have unit norm - direction matters more than magnitude.
Min-Max Scaling	Rescales values to a fixed range, typically [0, 1]: $(x - \text{min}) / (\text{max} - \text{min})$.
MaxAbs Scaling	Scales each feature by its maximum absolute value, preserving sign and sparsity, range [-1, 1].
Robust Scaling	Uses the median and IQR instead of mean/std, making it resistant to outliers.

Part G: Feature Construction & Transformation

Concept	Definition
FunctionTransformer	Wraps any custom function (e.g. log, sqrt, reciprocal) as a scikit-learn transformer.
PowerTransformer (Box-Cox / Yeo-Johnson)	Applies a power transform to make skewed data more Gaussian-like. Box-Cox requires strictly positive values; Yeo-Johnson supports zero/negative values too.

ColumnTransformer	Applies different preprocessing pipelines to different subsets of columns (e.g. scaling for numeric, encoding for categorical) within a single object.
Feature Construction	Creating new, more informative columns from existing ones - e.g. Debt-to-Income ratio = $\text{loan_amount} / \text{annual_income}$.

Part H: Final Deliverable

A **fully cleaned, encoded, scaled, and feature-engineered dataset**, documented with a project report covering the missing-value strategy, outlier treatment, encoding/scaling choices, engineered features, and final dataset readiness for Machine Learning modeling.